

Smoke the knives.

Do not reprint Fig. 5.

Ablation Bot AUTHORITATIVE ranks 1–7 (re-aligned over DELIVERABLE pending note). Primary: RC1 FVLMoE vs π_0 , RC2 fusion, RC3 insertion metrics. Smoke $\neq$ paper claims.

#1	#2	#3	#4	#5	#6	#7
Tab.3 fusion	Insert Plug	Insert USB	Averages	Occlusion	Multi-task	Pump caveat
Y	Y	Y	Y	Part	Part	cite

Pass: directional morals on fusion + insertion with force_fusion logged.

Fail: writing “we reproduced Fig.5 60.5%” off a 1-task stub. N: paper-scale Flexiv / 8×4090.

Five facts that do not change with the seat

01

Late 6-axis F/T + FVLMoE

After VLM (PaliGemma/SigLIP): project wrench → force token → FVLMoE (E=4, top-1) → additive guide into π_0 flow-matching action head. Force stays late.

02

Contact-rich on Flexiv

Insert USB / plug, pump bottle, wipe board, peel cucumber. ForceVLA-Data: 244 traj / ~140k steps / ~50 demos per task. Quest3 teleop.

03

Must-cites

Avg 60.5 vs π_0 -base w/o F 37.3 (+23.2). Naive w/ F only 40.2 (+2.9). Early MoE-before-VLM 0% → late FVLMoE 80% (Tab.3). Occlusion 90%. FAST+naive force 31.0→14.2.

04

CLEAR contrasts

NOT MoPA-PD ancestor (MoPA = MP privilege→vision student; ForceVLA = deploy force into VLA). NOT JEPA/WM child. No $\pi_{0.5}$ bake-off — do not invent.

05

Number hygiene

Typo fix: 37.3 = w/o F (not w/ F). Preserve paper numbers. No invented BEHAVIOR SR. Pump caveat: naive w/F 83 > ForceVLA 67 — do not overclaim.

AUTHORITATIVE — fold over DELIVERABLE pending

1	Tab.3 fusion-stage	Y	Early MoE 0% / concat 60 / FVLMoE 80. Cleanest FVLMoE test.
2	Fig.5 Insert Plug	Y	fast 25/30; base 45/60; ForceVLA 80. Abstract 80% story.
3	Fig.5 Insert USB	Y	Force on/off NULL for π_0 (0/0, 5/5); FVLMoE only gain $\rightarrow$ 25.
4	Fig.5 averages + FAST hurt	Y	60.5 vs 37.3 (+23.2); naive +2.9; fast 31.0 $\rightarrow$ 14.2.
5	Tab.2 occl / Obj Gen.2	Part	Occlusion 90 vs 60/30; Obj Gen.2 40. Naive can hurt.
6	Tab.5 multi-task	Part	Avg 67.5 vs 42.5 / 5 / 0. Plug saturates both force methods.
7	Pump Bottle reversal	cite	Naive w/F 83 > ForceVLA 67. Falsifier — do not overclaim.

Fusion stage: early MoE 0% vs late FVLMoE 80%

is, (7) Visual Occlusion, where parts of the plug and ; introducing physical instability via clutter beneath ial robustness and physical adaptability, with results

ior generalization, particularly in scenarios requiring it achieved an 80.00% success rate, outperforming 1 it naively. In the Height Gen. setting, ForceVLA able depths, avoiding torque limit violations seen in ned high success under visual degradation (90.00% in ultimodal feedback beyond visual cues. These results /LMoE architecture in intelligently integrating force r modulating action in response to dynamic physical t robotic manipulation.

LA, partic-
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Model	Success Rate
baseline 10	45%
linear before VLM	55%
MoE before VLM	0
concat after VLM	60%
ForceVLA (Ours)	80%

#1 / RC2 smoke

{early_linear, early_moe_stub, late_concat, late_fvlmoe} on 1 task / short FT.

Hard log

force_fusion = early_linear | early_moe | concat | fvlmoe.
force_source = true|est|stub.

Paper moral

Late ≥ early. Early MoE must not win (cite 0% if untrainable).

Not required

Matched Tab.3 80% absolute. Task identity for Tab.3 is underspecified — treat as plug-scale correspondence.

Insert Plug 80% story + USB force-null

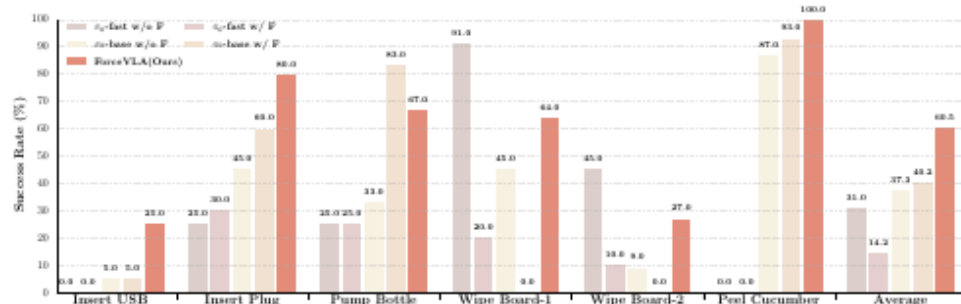


Figure 5: Main task success rates across different methods. ForceVLA significantly outperforms all baselines on five contact-rich tasks. Incorporating external force feedback improves performance for π_0 -base model, while our method achieves the highest average success rate, demonstrating robust performance under complex interaction dynamics. “Wipe Board-1” indicates the success rate of successfully performing the wiping motion, while “Wipe Board-2” refers to the success rate of completely erasing the markings.

ForceVLA shows an improvement of 23.2%. This highlights the substantial benefit of incorporating and effectively processing multimodal information, including force. Table 1 further highlights ForceVLA’s superior performance on the intricate *cucumber peeling* task. Our model excelled on both key metrics: it achieved the longest average peel length per stroke (14.12 cm ↑), indicating better ability to execute high-fidelity surface manipulation through stable tool orientation, adaptive contouring, and sustained surface contact compared to both π_0 -base w/o F (13.17 cm) and π_0 -base w/ F (13.17 cm).

#2 Insert Plug

25/30 → 45/60 → 80

Naive +15 on base; FVLMoE +20 more.

#3 Insert USB

Force on/off NULL for π_0

fast 0/0; base 5/5; FVLMoE 25 only gain.

RC3 harness

Log SR, align error, peak |F|. Prefer Plug; USB shows modality≠gain.

Averages +23.2 and the FAST hurt

π_0 -fast w/ F

14.2%

naive force HURTS

π_0 -fast w/o F

31.0%

FAST foil

π_0 -base w/o F

37.3%

headline foil (typo: not w/ F)

π_0 -base w/ F

40.2%

+2.9 only

ForceVLA

60.5%

+23.2 vs 37.3

Occlusion Partial · multi-task Partial/cite

Figure 6: Variants of generalization settings used in our experiments. (a–b) Different object geometries; (c) variation in socket height; (d) partial visual occlusion; (e) unstable socket conditions. These scenarios evaluate robustness under diverse physical and perceptual perturbations.

Table 2: Success rates (%) of different models under various experimental conditions. Maximum values in each column are highlighted in **bold**; second-best values are underlined.

Model	Object Gen. 1	Object Gen. 2	Height Gen.	Visual Occlusion	Unstable Socket	Average
π_0 -base [10] w/o F	48.00%	10.00%	66.67%	60.00%	10.00%	38.93%
π_0 -base [10] w/ F	32.00%	10.00%	77.78%	30.00%	10.00%	31.96%
π_0 -fast [25] w/o F	80.00%	<u>35.00%</u>	88.89%	50.00%	10.00%	52.78%
π_0 -fast [25] w/ F	32.00%	5.00%	44.44%	50.00%	30.00%	32.29%
ForceVLA (Ours)	80.00%	40.00%	88.89%	90.00%	<u>20.00%</u>	63.78%

height and measures success under torque limits. (d) **Visual Occlusion**, where parts of the plug and

frequency, reflects not only temporal semantic variance from tasks but also architectural bias toward certain experts, potentially shaped during training.

D Multi-task Evaluation

Table 5: Multi-task joint training success rates (%). ForceVLA (Ours) demonstrates superior average performance and excels or matches the best performance in all individual tasks. Best performance(s) in each column are in **bold**.

Model	Pump Bottle	Insert Plug	Insert USB	Wipe Board	Average SR
π_0 -fast w/o F	0.0%	0.0%	0.0%	0.0%	0.0%
π_0 -fast w/ F	0.0%	0.0%	0.0%	0.0%	0.0%
π_0 -base w/o F	20.0%	0.0%	0.0%	0.0%	5.0%
π_0 -base w/ F	50.0%	100.0%	10.0%	10.0%	42.5%
ForceVLA (Ours)	80.0%	100.0%	10.0%	80.0%	67.5%

#5 RC5 Partial

Occlusion stub: mask socket/plug ROI. Force-on $\geq$ force-off directional.

Paper: ForceVLA 90%; base w/F 30% (naive hurts).

#6 Tab.5 Partial/cite

Optional 2-task short joint FT; else cite 67.5 / 42.5 / 5 / 0.

Plug 100 both force methods — does not isolate FVLMoE.
USB 10% both — soft precision insert.

Pump Bottle: naive w/F 83 > ForceVLA 67

Falsifier column

Fig.5 Pump: fast 25/25; base w/o F 33; base w/F 83; ForceVLA 67. Naive state-concat beats FVLMoE by 16 pp.

Replication order

Replicate Plug/USB first. Treat Pump as a falsifier, not a confirmation. Do not bury it in speaker notes only.

Tab.5 regime shift

Under joint train Pump flips: ForceVLA 80 > base w/F 50. Regime matters — log training regime.

Honesty rule

Never brief 'ForceVLA wins every contact task.' Avg +23.2 and Plug 80 survive; Pump does not.

RC0–RC7 · what Y / Partial / N mean

RC0	data ingest	Y	Schema {V ^b ,V ^h ,s,f∈R ⁶ ,a,L}; wrench dim=6; <1h
RC1	FVLMoE vs π _o	Y	3 methods × 1 task short; log force_fusion
RC2	fusion stage	Y	early vs late; early_moe must not win
RC3	insertion metrics	Y	SR / align / peak F harness; no invented scores
RC4	data scale	Y/P	{10,25,50} demos stub or coverage report
RC5	occlusion gen	Part	mask ROI; directional force-on ≥ off
RC6	router log	Part	top-1 hist; non-degenerate only
RC7	multi-task	cite	optional 2-task stub; else Tab.5 cite
—	paper Flexiv	N	8×4090 / 5-task 20-trial protocol out of course

Smoke hygiene — refuse these sentences

No Fig.5 laundry

Do not claim 60.5% absolute from 1-task Partial smoke. Label smoke SR as smoke.

No Tab.3 laundry

Do not claim 80% without matched task + fusion settings. Cite 0% if early MoE untrainable.

No stub = ForceVLA-Data

Sim/stub wrench $\neq$ ForceVLA-Data reproduction. Log force_source honestly.

No FAST+F substitute

π_0 -fast + naive F regresses 31 $\rightarrow$ 14.2. Not a fair FVLMoE stand-in.

No paper-scale as smoke

8 $\times$ 4090 / 30k multi-task / full 5-task Flexiv eval = N for course.

No invented bars / $\pi_{0.5}$

Quote prose/tables/printed labels. No $\pi_{0.5}$ numbers. No invented BEHAVIOR SR.

Do not round away — refuse laundering foils

Averages

ForceVLA 60.5; base w/o F 37.3; w/F 40.2;
fast w/o F 31.0; w/F 14.2; Δ +23.2

Insertion

Plug 80; USB 25 (π_0 force-null). Peel 14.12 cm
/ 7 strokes.

Ablation / gen

Tab.3: 45 / 55 / 0 / 60 / 80. Occlusion 90. Gen
avg 63.78.

Data / train

244 traj / 140k / ~50/task. LR $2.5e-5 \rightarrow 2.5e-6$.
Multi 30k/~12h 2×4090 . Single 10k/~9h.

Hyperparams

Force 6 $\rightarrow$ 2048; MoE D=2048, E=4, k=1, 8
heads; action expert 1024; Adam β 0.9/0.95;
bf16; clip 1.0

One-liner

Quote 60.5/37.3/+23.2, occlusion 90, early
MoE 0 vs late 80; refuse FAST or $\pi_{0.5}$ as
headline foil.

Directional morals. Honest force_source. No Fig.5 reprint.

Minimum: RC1 + RC2 + RC3 with Ablation Bot morals #1–#4.

Cite Pump #7. Partial occlusion/router. N paper-scale Flexiv.

If someone pastes smoke SR into a Fig.5 cell — that run fails Empiricist review.

Next: Archaeologist — π_0 parent → late force → FVLMoE → FM; CLEAR vs MoPA / JEPA.